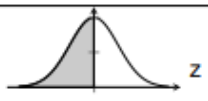
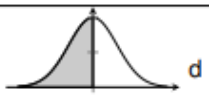
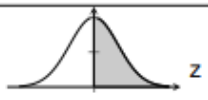
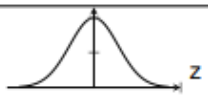
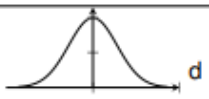
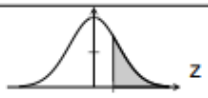
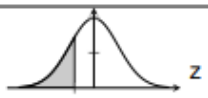
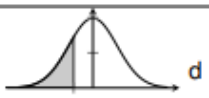
The Normal Distribution

LLE – Mathematics and Statistics

1. These questions concern the standard normal distribution, Z, which has a mean $\mu = 0$ and a variance $\sigma^2 = 1$, so you can write $Z \sim N(0,1)$. You will also consider the distribution of the length of donkey's tails, $D \sim N(35,16)$, that is the normal distribution with mean $\mu = 35$ cm and variance $\sigma^2 = 16$ cm^2. Fill in the blanks using the information throughout the table below. No tables or calculators are required.

$P(Z<0) =$ 0.5		$P(D<35cm) =$ 0.5	
$P(Z>0) =$ 0.5		$P(D>35cm) =$ 0.5	
$P(Z>-4) \approx$ 1		$P(D>19) \approx$ 1	
$P(Z>4) \approx$ 0		$P(D>51) \approx$ 0	
$P(Z<0.8) =$ 0.788		$P(D<38.2) =$ $= P(Z<0.8) = 0.788$	
$P(Z>0.8) =$ 0.212		$P(D>38.2) =$ 0.212	
$P(Z<-0.8) =$ 0.212		$P(D<31.8) =$ 0.212	
$P(-0.8<Z<0.8) =$ $0.788-0.212=0.576$		$P(31.8<D<38.2) =$ 0.576	
$P(Z>-0.8) =$ 0.788		$P(D>31.8) =$ 0.788	

2. Calculate the standardised score z and the non-standardised score x for each of the distributions below

Distribution for X	x**-value (raw value)**	z**-value (standardised value)**
$X \sim N(30, 25)$	35	1
$X \sim N(20, 25)$	20	-2
$X \sim N(100, 256)$	124	1.5
$X \sim N(100, 256)$	140	2.5
$X \sim N(50, 6^2)$	29	-3.5

3. Given the information about the population distribution, find the probabilities requested by first standardising the raw scores.

   (a) $X \sim N(80, 25)$

   i. $P(X < 85) = P(Z < \frac{85-80}{5}) = P(Z < 1) = 0.8413$

   ii. $P(X > 70) = P(Z > -2) = 0.9773$

   iii. $P(78 < X < 82) = P(-0.4 < Z < 0.4) = 0.3108$

   (b) $X \sim N(150, 156.25)$

   i. $P(X < 143.75) = P(Z < \frac{143.75-150}{12.5}) = P(Z < -0.5) = 0.3085$

   ii. $P(X > 180) = P(Z > 2.4) = 0.0082$

   iii. $P(145 < X < 170) = P(-0.4 < Z < 1.6) = P(Z < 1.6) - P(Z < -0.4) = 0.6096$

4. Given the information about the population distribution, find the values that are represented by the following probabilities/proportions. Give your answers correct to 1 decimal place. You will need to find the z-score from the probability given and then transform this back to a raw score.

   (a) $X \sim N(100, 16)$

   i. $P(X < r) = 0.9772, z = 2, r = 100 + 2 \times 4 = 108$

   ii. $P(X > s) = 0.9772, z = -2, s = 92$

   iii. $P(X < t) = 0.35, z = -0.385, t = 98.5$

(b) $X \sim N(20, 4)$

i. $P(X < r) = 0.9772, z = 2, r - 20 + 2 \times 2 = 24$

ii. $P(X > t) = 65\%, z = -0.385, t = 19.2$

5. A hot drinks machine is known to dispense liquid in a volume that is normally distributed with a mean of 250 ml and a standard deviation of 5 ml, $V \sim N(250, 25)$

   (a) What proportion of drinks are expected to be dispensed that are over 254 ml?
   $z = \frac{254-250}{5} = 0.8$
   $P(V > 254) = P(Z > 0.8) = 0.2119 = 21.2\%$

   (b) What proportion of drinks that are expected to be under 246 ml in volume.
   $P(V < 246) = P(Z < -0.8) = 0.2119 = 21.2\%$

   (c) Use your answers and sketches to parts b and c to work out the probability that a drink is dispensed that is between 246 ml and 254 ml.
   Over 254 ml and under 246 ml is 21.2 + 21.2 = 42.4%
   Therefore between these amounts is 100 - 42.4 = 57.6 %

   (d) What volume are 75% of drinks less than?
   $P(V < v) = 0.75$
   $P(Z < 0.674) = 0.75 \implies v = 250 + 0.674 \times 5 = 253.4$ ml

   (e) You find that your cup overflows 2% of the time. How big is your cup?
   $P(V > c) = 0.02$
   $P(Z > 2.054) = 0.02 \implies c = 250 + 2.054 \times 5 = 260.3$ ml